3.1.1 Particle in a confining potential

Before proceeding to describe the quantum harmonic oscillator, let us first recall what happens in the case of a particle moving in a general potential. If the potential is constant, such that the particle is freely moving, then its energies will form a continuum. However, if the potential is sufficiently confining then energy becomes quantised. By trapping the particle, discreteness appears. For example, the attractive potential from a proton experienced by an electron in a hydrogen atom leads to the formation of a discrete set of eigenstates (orbitals) and associated distinct energies.

In general, the energy eigenvalues and eigenstates for a given physical system are determined by the system Hamiltonian $\hat{H}$. They are found by solving the time-independent Schrödinger equation

$$\hat{H}|\psi\rangle = E|\psi\rangle \tag{3.1}$$

i.e. the eigenequation for $\hat{H}$. For a single, nonrelativistic particle of mass m moving in one dimension in a potential $V(x)$, the Hamiltonian is given by

$$\hat{H} = \frac{\hat{p}^2}{2m} + V(\hat{x}), \tag{3.2}$$

where $\hat{x}$ and $\hat{p}$ are the observables associated with position and momentum, respectively. They obey the canonical commutation relation[1]

$$[\hat{x}, \hat{p}] = i\hbar. \tag{3.3}$$

Working in the position eigenbasis, we can represent states by their position-space wavefunctions, $\psi(x) = \langle x|\psi\rangle$. The action of the momentum operator becomes

$$\langle x|\hat{p}|\psi\rangle = -i\hbar\frac{\mathrm{d}}{\mathrm{d}x}\psi(x), \tag{3.4}$$

and for the potential we have $\langle x|V(\hat{x})|\psi\rangle = V(x)\psi(x)$. The Schrödinger equation in position space is therefore

$$\left[-\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}x^2} + V(x)\right]\psi(x) = E\,\psi(x). \tag{3.5}$$

This is a second-order differential equation, which should be solved to find the energy eigenvalues and -states of the system. Example 3.1 gives the solution for the simple case of a particle in a one-dimensional box.

Example 3.1: Particle in a box

A simple case, for which the Schrödinger equation can be solved exactly, is the somewhat artificial but illustrative infinite-square-well potential

$$V(x) = \begin{cases} 0 & \text{for } 0 \leq x \leq L \\ \infty & \text{otherwise} \end{cases}. \tag{3.6}$$

[1] Note that, while this notation is convenient and commonplace, strictly speaking, an operator cannot equal a number. The right-hand side in (3.3) should really be understood as multiplying the identity operator $i\hbar\mathbb{1}$. Similarly for other constant operators.

The particle is confined to a region of length L by infinite potential walls.

Outside the well, the wavefunction must vanish (otherwise the energy will diverge). Inside the well, the Schrödinger equation gives

$$\frac{\mathrm{d}^2}{\mathrm{d}x^2}\psi(x) = -\frac{2m}{\hbar^2}E\,\psi(x) = -\kappa^2\,\psi(x), \tag{3.7}$$

where $\kappa^2 = 2mE/\hbar^2$. This linear, second-order, homogeneous differential equation has a general solution of the form

$$\psi(x) = Ae^{-i\kappa x} + Be^{i\kappa x}, \tag{3.8}$$

where the coefficients A, B must be determined from boundary conditions. The wavefunction must vanish at the edges of the well (as it is zero outside and must be continuous). Hence

$$\psi(0) = A + B = 0, \tag{3.9}$$

so $\psi(x) = A(e^{-i\kappa x} - e^{i\kappa x}) = 2iA\sin(\kappa x)$, and

$$\psi(L) = 2iA\sin(\kappa L) = 0. \tag{3.10}$$

This implies that κL must be an integer multiple of π. The eigenenergies are therefore

$$E_n = \frac{\pi^2\hbar^2}{2mL^2}n^2 = \frac{h^2}{8mL^2}n^2, \qquad n = 0, 1, 2, \ldots. \tag{3.11}$$

Furthermore, normalisation requires that

$$1 = \int_0^L |\psi(x)|^2 dx = \int_0^L 4|A|^2\sin^2(\frac{n\pi}{L}x)dx = 2|A|^2 L, \tag{3.12}$$

which implies $|A| = 1/\sqrt{2L}$. Note that the phase of A can be chosen arbitrarily since it has no physical significance. We can thus take the energy eigenstate wavefunctions to be real

$$\psi_n(x) = \sqrt{\frac{2}{L}}\sin\Big(\frac{n\pi}{L}x\Big), \qquad n = 0, 1, 2, \ldots. \tag{3.13}$$

They are standing waves in the well, fitting an integer number of half wavelengths between the two walls.

The solutions for the energy eigenvalues and eigenstates can be found for a variety of simple potentials beyond the infinite square well. Examples include the finite-depth square well, the delta-function potential, and the step potential. However, all these potentials are somewhat artificial, and although some of them are good approximations to real-life systems, they are mainly chosen for the simplicity of finding analytical solutions. Real physical potentials are often too complicated to allow for analytical solutions. One example is the potential experienced by trapped electrons in a Nitrogen-Vacancy center in

diamond (an important system for quantum sensing discussed in Chap. 6). The solution to this complicated potential can only be found using sophisticated numerical methods, which yield approximate solutions. Nevertheless, there are a few simple potentials which both describe realistic physical systems very well and are amenable to analytical solution. One example is the Coulomb potential, which accurately describes the potential experienced by an electron in a hydrogen atom. Another important example is the harmonic potential, which we will now discuss.

3.1.2 Harmonic potential

A harmonic potential with angular frequency ω for a particle of mass m moving in one dimension is given by

$$V(x) = \frac{1}{2}m\omega^2 x^2, \tag{3.14}$$

and the Hamiltonian (3.2) is $\hat{H} = \frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2\hat{x}^2$. It is often convenient to rescale the position and momentum operators,

$$\hat{x} \to \sqrt{\frac{m\omega}{\hbar}}\,\hat{x} \quad \text{and} \quad \hat{p} \to \frac{1}{\sqrt{m\omega\hbar}}\,\hat{p}. \tag{3.15}$$

The commutator of the rescaled operators is then

$$[\hat{x}, \hat{p}] = i, \tag{3.16}$$

and the Hamiltonian takes the form

$$\hat{H} = \frac{1}{2}\hbar\omega(\hat{x}^2 + \hat{p}^2). \tag{3.17}$$

This harmonic oscillator Hamiltonian can represent a large number of different physical systems used in quantum information technology. They include, for example, the electromagnetic field (particularly optical waves and microwaves), the inductor-capacitor electronic circuit, acoustic sound waves in bulk materials, and mechanical oscillators such as oscillating membranes or cantilevers. Despite the apparent differences in these systems, they can all be described by the same Hamiltonian and thus the same mathematics. However, the'position' and 'momentum' operators $\hat{x}$, $\hat{p}$ represent different quantities depending on the problem. For mechanical oscillators, they represent actual position and momentum, but for the electromagnetic field, $\hat{x}$ and $\hat{p}$ represent the amplitude and phase quadratures of the field (as will be discussed below), while for inductor-capacitor electronic circuits, they represent charge and current.

To find the energy eigenvalues and eigenstates for the quantum harmonic oscillator, we must solve the time-independent Schrödinger equation. However, for this system there exists a much more elegant approach than solving differential equation for the wavefunction directly. We start by introducing the operators

$$\hat{a} = \frac{1}{\sqrt{2}}(\hat{x} + i\hat{p}) \quad \text{and} \quad \hat{a}^\dagger = \frac{1}{\sqrt{2}}(\hat{x} - i\hat{p}). \tag{3.18}$$